Amazon API Gateway

What is Amazon API Gateway?

Amazon API Gateway is an AWS service for creating, publishing, maintaining, monitoring, and securing REST [Representational State Transfer (REST) architectural style] HTTP, and WebSocket APIs at any scale. API developers can create APIs that access AWS or other web services, as well as data stored in the AWS Cloud. **As an API Gateway API developer, you can create APIs for use in your own client applications. Or you can make your APIs available to third-party app developers.**

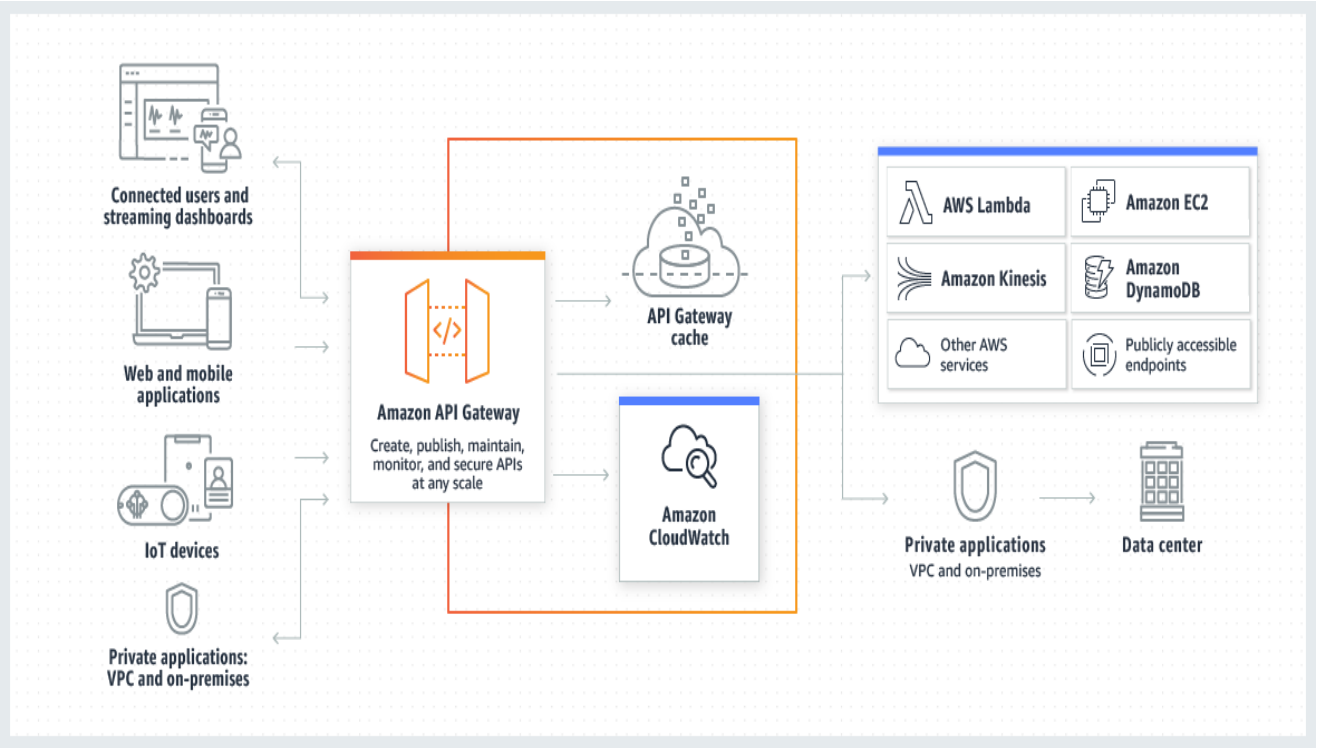
API Gateway creates RESTful APIs that:

- Are HTTP-based.
- Enable stateless client-server communication.
- Implement standard HTTP methods such as GET, POST, PUT, PATCH, and DELETE.

API Gateway creates WebSocket APIs that:

- Adhere to the WebSocket protocol, which enables stateful, full-duplex communication between client and server.
- Route incoming messages based on message content.

Architecture of API Gateway



This diagram illustrates how the APIs you build in Amazon API Gateway provide you or your developer customers with an integrated and consistent developer experience for building AWS serverless applications. **API Gateway handles all the tasks involved in accepting and processing up to hundreds of thousands of concurrent API calls. These tasks include traffic management, authorization and access control, monitoring, and API version management.**

API Gateway acts as a "front door" for applications to access data, business logic, or functionality from your backend services, such as workloads running on Amazon Elastic Compute Cloud (Amazon EC2), code running on AWS Lambda, any web application, or real-time communication applications.

Features of API Gateway

Amazon API Gateway offers features such as the following:

- Support for stateful (WebSocket) and stateless (HTTP and REST) APIs.
- Powerful, flexible authentication mechanisms, such as AWS Identity and Access Management policies, Lambda authorizer functions, and Amazon Cognito user pools.
- Canary release deployments for safely rolling out changes.
- CloudTrail logging and monitoring of API usage and API changes.
- CloudWatch access logging and execution logging, including the ability to set alarms.
- Ability to use AWS CloudFormation templates to enable API creation
- Support for custom domain names.
- Integration with AWS WAF for protecting your APIs against common web exploits.
- Integration with AWS X-Ray for understanding and triaging performance latencies.